

# Yashas Kadambi

M.S. Computer Science, UC San Diego | Prior Teaching Assistant | Software Engineer

ykadambi@ucsd.edu | LinkedIn | GitHub | Website | Teaching Demo

## TEACHING PROFILE

Prior TA experience in Image Processing and Computer Vision, Data Analytics, and Deep Learning Theory and Practices, supporting 656 students across three courses. Designed assignments, hands-on labs, answer keys, solution notebooks, tutorial videos, and lab demonstrations; built grading workflows for assignments and competitions; and authored three graduate-course assignments. Also brings 3+ years of software engineering experience at Cisco and two peer-reviewed publications.

## EDUCATION

### University of California San Diego

Expected 2027

M.S. in Computer Science | San Diego, California

### PES University

Aug 2019 - May 2023

B.Tech. in Computer Science and Engineering | TEC-evaluated U.S. GPA: 3.78/4.00 (124.5 semester credits) | MRD Scholarship

## TEACHING EXPERIENCE

### PES University | Teaching Assistant

Jul 2022 - May 2023

Department of Computer Science and Engineering | Bengaluru, India

### Image Processing and Computer Vision | 122 students

Dec 2022 - May 2023

Instructor: Prof. Gowri Srinivasa

- Designed a MATLAB-based steganography assignment and hands-on lab; created [tutorial videos and led lab demonstrations](#) with guided examples covering image sampling, intensity transformations, histogram equalization, ROI masking, and LSB watermarking; authored the complete answer key.
- Graded approximately 250 submissions within one week; answered approximately 50 student questions and held five 2-hour office-hour sessions.
- Built a reusable grading macro that automated repetitive checks and reduced assignment-evaluation time by 50%.
- Tech stack:** MATLAB and Image Processing Toolbox.

### Data Analytics | 494 students

Aug 2022 - Dec 2022

Instructor: Prof. Gowri Srinivasa

- Designed a Brooklyn Nine-Nine-themed assignment and public Kaggle team competition to teach Markov chains, absorbing states, stationary distributions, and A/B testing; created [tutorial videos and led lab demonstrations](#) with worked examples; authored the unit materials, solution notebook, and answer key.
- Manually graded approximately 370 assignment submissions within one week; answered approximately 150 questions through six 2-hour office-hour sessions and course forums.
- Built the competition's automated grader and manually verified submissions from 178 teams, returning consistent results within one day.
- Tech stack/platforms:** Python, NumPy, pandas, Matplotlib, scikit-learn, Jupyter/Google Colab, and Kaggle.

### Deep Learning Theory and Practices [Graduate Course] | ~40 students

Jul 2022 - Dec 2022

Instructors: Prof. Preet Kanwal and Prof. Uma P

- Selected as an undergraduate TA for the ~40-student graduate course Deep Learning Theory and Practices; authored three assignments, hands-on labs, and teaching materials on autoencoders and generative adversarial networks.
- Developed an MNIST autoencoder notebook covering encoder-decoder construction, custom training, image reconstruction, and TensorBoard visualization.
- Built a web-based submission and autograding workflow that returned instant validation for all three assignments; supported students through two dedicated office-hour sessions and hands-on debugging.
- Tech stack:** Python, NumPy, pandas, TensorFlow/Keras, PyTorch, and TensorBoard.

## TEACHING AND TECHNICAL SKILLS

### Teaching & Assessment

Assignment/lab design; tutorial videos and lab demonstrations; answer keys and solution notebooks; Kaggle competition hosting; grading and feedback; office hours/forums; autograding; mentoring

### Programming & Machine Learning

Python; MATLAB; C++; Java; SQL; NumPy; pandas; SciPy; Matplotlib; scikit-learn; OpenCV; TensorFlow/Keras; PyTorch; Jupyter; TensorBoard

### Systems & Cloud

Git; Linux; AWS; Terraform; Docker; CI/CD; PostgreSQL; MongoDB; Cassandra; REST APIs

## PROFESSIONAL EXPERIENCE

### Cisco

Jan 2023 - Present

Cloud platforms and optical networking | Bengaluru, India

#### Optical Software Development Engineer II

Jan 2025 - Present

- Designed and implemented a comprehensive unit-testing framework that reduced testing time for new features by more than 90%, improving release velocity and test consistency.
- Developed end-to-end data-plane software for NCS 1014/Aquila optical line cards, including CDR hardware integration, secure boot, and QPSK modulation for greater reach.
- Created internal tools for automated builds and deployments, near-real-time device-log streaming, and performance-monitoring extraction to accelerate debugging and developer workflows.

#### Software Engineer

Aug 2023 - Jan 2025

- Developed reusable Terraform modules for AWS infrastructure and parallelized deployment plans, cutting deployment time by 50% while improving consistency across services.
- Identified root causes of four production outages and resolved performance bottlenecks across PostgreSQL, MongoDB, and Cassandra, reducing page-load time by 40%.
- Mentored two new engineers through structured knowledge-transfer sessions, code reviews, code-quality expectations, and hands-on debugging.

#### Software Development Intern

Jan 2023 - Jul 2023

- Built a CI pipeline that generated and published Python and Java client SDKs on API changes and delivered an AWS Glue data-integration job for the PX Cloud platform.

#### Schneider Electric | Summer Intern

Jun 2022 - Aug 2022

- Independently designed and built a software tool that reduced switchgear test-case discussion time by more than 90%, from two days to two hours.

## ACADEMIC AND RESEARCH PROJECTS

- SWIFT - Lightweight Image Super-Resolution:** Co-developed the official PyTorch implementation of a SwinV2 Transformer and Fast Fourier Convolution model, including training, benchmarking, Docker, and TorchServe inference.
- Multi-view 3D Reconstruction:** Implemented an incremental Structure from Motion pipeline for sparse 3D point-cloud reconstruction using SIFT, pose estimation, triangulation, and bundle adjustment.
- Chocollvm Compiler Frontend:** Collaborated on a ChocoPy-subset compiler frontend that produces type-checked ASTs and LLVM IR, with a CLI and automated tests.
- SSML - Spark Streaming for Machine Learning:** Collaborated on distributed CIFAR-10 image classification using Apache Spark/PySpark, TensorFlow/Keras, and Kafka for a Big Data course project.
- Bitcoin Protocol from Scratch:** Built SHA-256, RIPEMD-160, secp256k1 operations, wallet generation, transaction signing, and a Bitcoin testnet transaction without external dependencies.
- Restaurant Analysis using the Yelp Dataset:** Used Yelp check-in and review data for restaurant closure-risk analysis, exploratory analysis, classification, and LDA-based feature discovery.
- Systems and Parallel Programming Coursework:** C and Python assignments covering multithreaded Pi computation, thread-error analysis, and parallel-programming exercises.
- Cloud Provisioning using RDBMS:** Developed a database-backed application for allocating cloud hardware to users and tracking assignments, supported by ER modeling and full-stack components.
- Cloud-Hack: Kubernetes Microservices:** Deployed a containerized Flask/MongoDB blog as Kubernetes microservices using Deployments, Services, ConfigMaps, and Secrets.
- PeTra - Pet Travel and Sitter Booking:** Equal-contribution React application for booking pet-friendly hotels and pet sitters, developed for Web Technologies I.

## PUBLICATIONS

- [1] Co-author, "Toward Faster and Efficient Lightweight Image Super-Resolution Using Transformers and Fourier Convolutions," Artificial Intelligence and Applications, 3(2), 168-178, 2025.
- [2] Co-author, "Monitoring and Alert Systems for Underwater Data Centers using Arduino," 2021 IEEE International Conference on Computation System and Information Technology for Sustainable Solutions (CSITSS), pp. 1-6.

## HONORS AND CERTIFICATIONS

|                                             |                                          |                                                                        |
|---------------------------------------------|------------------------------------------|------------------------------------------------------------------------|
| <b>AWS Certified Developer</b><br>Associate | <b>MRD Scholarship</b><br>PES University | <b>Web Application Security CTF</b><br>Cisco organization-level winner |
|---------------------------------------------|------------------------------------------|------------------------------------------------------------------------|